

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

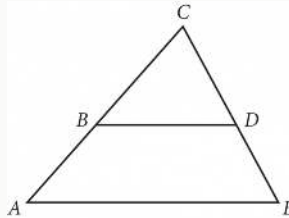
Geometry and Trigonometry

01

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

In the figure above, segments AE and BD are parallel. If angle BDC measures 58° and angle ACE measures 62° , what is the measure of angle CAE ?

- A. 58°
- B. 60°
- C. 62°
- D. 120°

Square X has a side length of 12 centimeters. The perimeter of square Y is 2 times the perimeter of square X. What is the length, in centimeters, of one side of square Y?

- A. 6
- B. 10
- C. 14
- D. 24

Q3

GRID-IN

GEOMETRY AND TRIGONOMETRY

What is the radius of the circle in the xy -plane defined by $x^2 + y^2 = 169$?

STUDENT-PRODUCED RESPONSE

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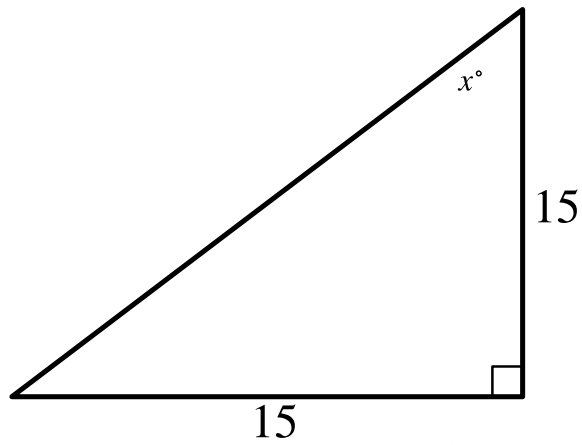
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The lengths of the 2 sides adjacent to the right angle are as follows:
 - 15
 - 15
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

In triangle ABC, the measure of angle A is 50° . If triangle ABC is isosceles, which of the following is NOT a possible measure of angle B ?

- A. 50°
- B. 65°
- C. 80°
- D. 100°

Q6

GEOMETRY AND TRIGONOMETRY

The length of one side of square M is 6 times the length of one side of square N. The area of square N is 49 square feet. What is the area, in square feet, of square M?

- A. 1,764
- B. 504
- C. 294
- D. 252

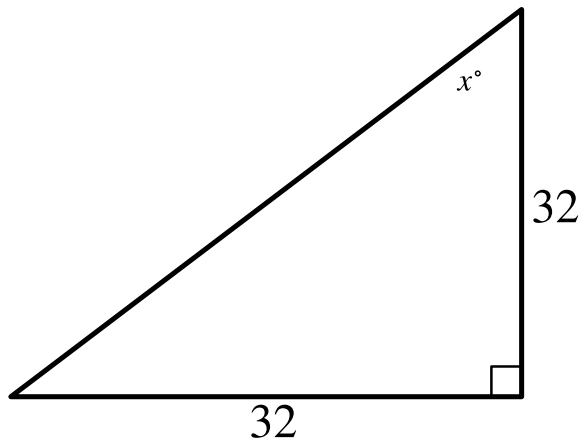
The measure of angle Z is 60° . What is the measure, in radians, of angle Z ?

A. $\frac{1}{6}\pi$

B. $\frac{1}{3}\pi$

C. $\frac{2}{3}\pi$

D. 1π



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The lengths of the 2 sides adjacent to the right angle are as follows:
 - 32
 - 32
- Note: Figure not drawn to scale.

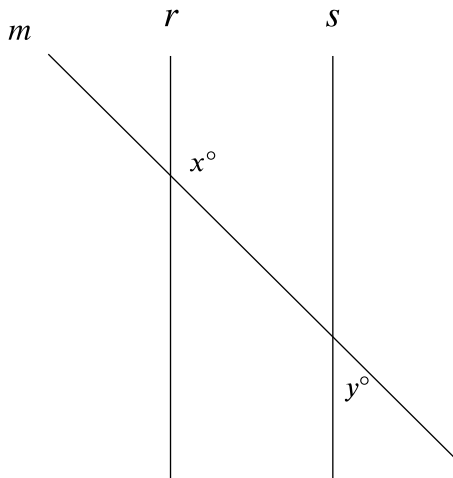
In the triangle shown, what is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

$\overline{9}$ $\overline{9}$ $\overline{9}$ $\overline{9}$

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Clockwise from top left, the 3 lines are labeled m , r , and s .
- Line m intersects both line r and line s .
- At the intersection of line m and line r , 1 angle is labeled clockwise from top left as follows:
 - Top right: x°
- At the intersection of line m and line s , 1 angle is labeled clockwise from top left as follows:
 - Bottom right: y°
- A note indicates the figure is not drawn to scale.

In the figure shown, lines r and s are parallel, and line m intersects both lines. If $y < 65$, which of the following must be true?

- A. $x < 115$
- B. $x > 115$
- C. $x + y < 180$

D. $x + y > 180$

A triangle has a base length of 10 centimeters and a corresponding height of 70 centimeters. What is the area, in square centimeters, of the triangle?

- A. 700
- B. 350
- C. 175
- D. 80